

Collaborative Reasoner : Self-Improving Social Agents with Synthetic Conversations

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With increasingly powerful large language models (LLMs) and LLM-based agents tackling an ever-growing list of tasks, we envision a future where numerous LLM agents work seamlessly with other AI agents or humans, facilitating everyday life in myriad ways – from problem-solving to planning, knowledge gathering, and learning. To navigate various scenarios in different social contexts, such LLM agents need to possess collaborative skills, such as effective communication and theory-of-mind. However, these skills are often overlooked in the predominant single-turn evaluation of LLMs. Moreover, improving the collaborative skills of these social agents requires large amounts of conversational data, which are often expensive and hard to collect. To bridge the gap between problem-solving and social collaboration skills of LLMs, we present **Collaborative Reasoner** (Coral ) , a framework to evaluate and improve the collaborative reasoning skills of language models. In particular, tasks and metrics in Coral necessitate agents to disagree with incorrect solutions, convince their partners of a correct solution, and ultimately agree as a team to commit to a final solution. Through the evaluation on five collaborative reasoning tasks, we show that current models cannot consistently utilize collaboration to achieve better task performance, and their social behaviors stemming from the current post-training process make them less desirable in collaborative scenarios. To improve the collaborative reasoning capabilities of LLMs, we propose a self-improvement approach using synthetic interaction data and to facilitate synthetic conversational data generation at scale, we built **Matrix**, a scalable and robust multi-agent communication framework. We leverage Coral and Matrix to synthesize supervised and preference fine-tuning data, based on the conversation turns that enable an agent to reach the correct answers. Experiments on general, math, scientific, and social reasoning tasks show the effectiveness of our approach, yielding improvements of up to 29.4% over the chain-of-thought performance of an equivalent single-agent LLM. We release the code for Coral and Matrix for future research on collaborative social agents.

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Collaborative Reasoner Code: <https://github.com/facebookresearch/collaborative-reasoner>

Matrix Code: <https://github.com/facebookresearch/matrix>



1 Introduction

Human collaboration often enables more effective and efficient solutions to difficult problems (Sampson and Clark, 2009; Woolley et al., 2010). Akin to such human teaming, our goal is to develop conversational agents that can collaborate with humans or other conversational agents to accomplish tasks better than what a single agent or human is able to do (Chen et al., 2023b; Feng et al., 2025). Imagine such an agent with the ability to partner with you on your graduate school homework or collaborate with you to practice for your upcoming job interview. These collaborations are challenging because in addition to problem-solving, they require social skills such as effective communication, providing feedback, having empathy and theory-of-mind, etc. Furthermore, such collaboration typically manifests over multiple turns of back-and-forth, natural conversation.

Instead of such collaborative and interactive problem-solving scenarios, modern-day LLMs are predominantly trained and evaluated for single-turn question-answering or problem-solving tasks (Gemini Team, 2023; OpenAI, 2023; Llama Team, 2024). Consequently, there is a gap in understanding how well current frontier language models can collaborate with other agents and humans in a natural, multi-turn conversation.

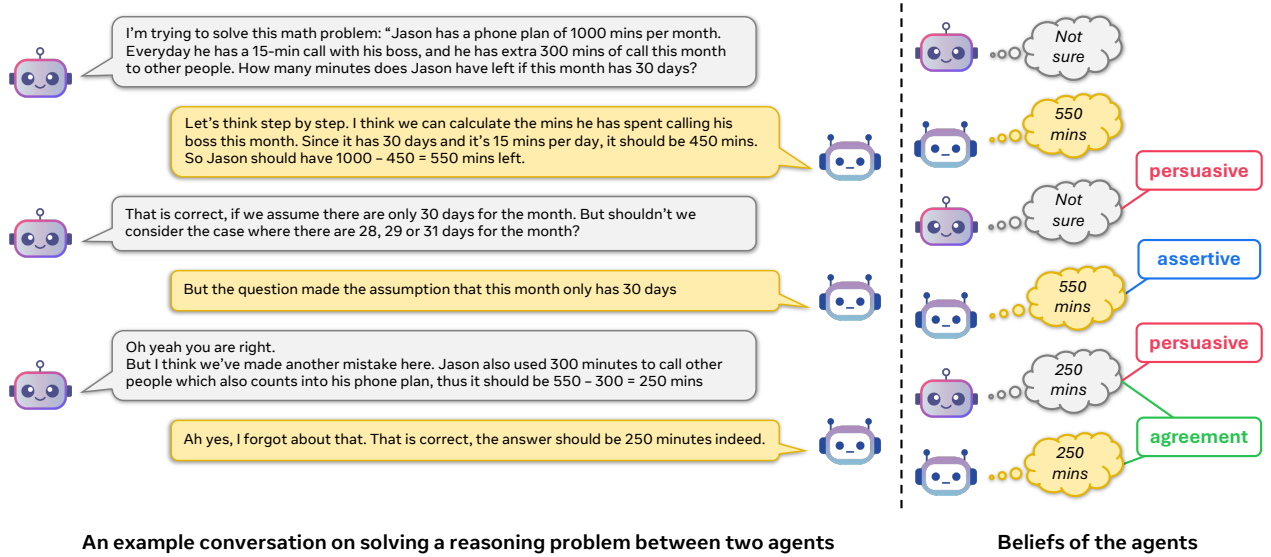


Figure 1 An example conversation for two agents to solve a reasoning problem collaboratively. Throughout the conversation, we track social behaviors such as *persuasion* and *agreement*, in addition to answer correctness.

Moreover, while multi-agent teams of LLMs with different roles, capabilities, and communication architectures have also been explored for complex problem-solving (Guo et al., 2024; Chen et al., 2023a; Zhu et al., 2025), little progress has been made in developing generalist agents that possess all the necessary reasoning and social skills for effective in-the-wild collaboration through natural conversations with humans (Zhao et al., 2024). On the other hand, developing such agents is currently challenging owing to lack of conversational collaboration data. Such data remains expensive to gather, and can be domain-specific and limited—making both training and evaluation difficult. Motivated by the lack of evaluation, training data, and training methods that can enable LLMs to collaborate over multiple turns of conversations in goal-oriented tasks, we present **Collaborative Reasoner** in this work, which we also refer as **Coral**¹.

Coral is a comprehensive framework focused on evaluating and improving the collaborative reasoning skills of language models. It entails a suite of goal-oriented tasks requiring multi-step reasoning that need to be accomplished collaboratively by two agents via a multi-turn conversation. More specifically, given a reasoning problem (e.g., math, physics), **Coral** emulates human-AI collaboration and requires two agents to work together on the problem over multiple turns. Along with solving the problem correctly, it also requires agents to agree with each other before committing to a final solution of a given problem. Consequently, learning to disagree to incorrect solutions *i.e.*, assertiveness, asking clarifying questions, and ability to convince the partner of a correct solution or persuasiveness, in addition to reasoning, are required to succeed. We evaluate several frontier open and closed sourced LLMs on 5 reasoning tasks under this collaborative setting, spanning domains across math, scientific question answering and social story comprehension. Compared with single-agent approaches such as chain-of-thought prompting, we find even these frontier models are inconsistent at leveraging collaboration to better approach these tasks. Further analysis on social metrics reveals a tendency for agents to be overly agreeable (> 90% agreement score), regardless of reasoning correctness, limiting their ability to challenge incorrect solutions and reducing collaboration efficacy.

To obtain large-scale, high-quality collaborative conversations in the absence of existing datasets, we propose to leverage synthetic conversations collected from simulated self-collaborations with an LLM itself. These synthetic conversations are later used to self-improve the LLM by training on the high-quality conversation turns after filtering. However, different from single-model and single-turn workflows, synthesizing conversational data at scale poses its own engineering challenges (e.g., optimize network flow). To that end, we build **Matrix**, a robust, versatile and high-performance model serving framework, which allows thousands of conversations being generated in parallel over hundreds of model instances. . With a large pool of collaborative conversational

¹Coral is short for collaborative reasoning models

data generated by **Matrix**, we employ both conversation-level and turn-level filtering methods to obtain supervised-finetuning or preference-finetuning data of for collaborative reasoners. Experiments show that this simple approach of sampling and filtering synthetic collaborative interactions, followed by supervised or preference-tuning improves the performance of **Llama-3.1** models by 9.0% to 29.5% across 8B and 70B model sizes. Our finetuned LLMs also outperform competitive single-agent baselines including the same base models finetuned on successful chain-of-thought reasoning traces, on all datasets. On theory-of-mind reasoning benchmark, our trained 70B collaborative reasoner also outperforms frontier models including GPT-4o and O1.

Our work paves the way for developing social agents that can partner with humans and other agents. We open-source our code for **Coral** and **Matrix** to support further research on such social agents. And we hope to extend our research to include evaluation and data collection of human-AI interactions to further foster research in this area.

2 Collaborative Reasoning over Multi-turn Conversations

The main goal for **Coral** is to develop collaborative capabilities of LLM agents, enabling them to partner with other agents over multiple turns of natural language conversations in solving various reasoning tasks. To that end, we first formally define the collaborative reasoning problem between two agents (humans or LLMs). We reformulate existing reasoning tasks that are mainly aimed towards single-agent, single-turn setting for multi-agent, multi-turn collaborative reasoning. Unlike single-turn setting where the performance only depends on the correctness of the generated solution, collaborative reasoning requires agents to agree on a solution as a team. We introduce metrics that capture such performance, in addition to social behaviors that manifest during collaboration such as assertiveness and persuasiveness.

2.1 Problem Definition

Given a reasoning problem $\{x, y^*\}$, where x denotes the task input (e.g., “Jason has a phone plan of ... How many minutes does Jason have left if this month has 30 days?” as in [Figure 1](#)) and y^* denotes the gold-standard task output (e.g., “250 mins”), collaborative reasoning entails two language model agents A and B engaging in a conversation to discuss how to solve this problem. To initiate the conversation, the first utterance a_1 is generated through a template of “I’m trying to solve this problem: [insert problem]”.² And the whole conversation consists of n rounds of interleaving utterances $C_n^u = \{a_1, b_1, a_2, b_2, \dots, u_n\}$ where $u \in \{a, b\}$ denotes the last speaker. More specifically, for each turn, agent u samples an utterance u_i based on the conversational prefix $C_{i-1}^{u'}$ and its system prompt s_u with language model P_u , i.e., $u_i \sim P_u(u_i | s_u, C_{i-1}^{u'})$. Here $C_{i-1}^{u'}$ represents the partial conversation history up till round $i - 1$ with the last speaker of u' , where $u, u' \in \{a, b\}, u \neq u'$. The conversation will stop when certain criteria is achieved or the maximum number of turns is reached. We use agreement between agents as the criteria for ending the conversation (e.g., [Figure 1](#)) and for evaluating the collaborative performance of agents on the problem.

2.2 Coral Metrics

Measuring agreement between agents on the solution to a given problem from agent generated utterances is challenging. When modeling reasoning problems in a single-turn, it is common to first generate a sequence u that represents the thinking process (e.g., chain-of-thought) followed by the final answer $\hat{y}$, i.e., $P_{\text{LM}}(\hat{y}|x) = P_{\text{LM}}(\hat{y}|u, x) * P_{\text{LM}}(u|x)$. However, in a multi-turn conversational setting, the end of each utterance may not necessarily mark the final answer itself, as the agents might be in the middle of working through the problem, or debating whose answer is correct. For example, a clarification question is asked in the third turn of the conversation in [Figure 1](#) which does not give any answers explicitly. We first propose a simple approach to extract implicit agent beliefs on the solution from their generated utterances. These beliefs not only enable computing conversation-level metrics such as agreement that serve as a measure of collaborative reasoning performance, but also allow computing turn-level metrics that capture social behaviors such as persuasion and assertiveness.

²Without loss of generality, we note A to be the agent that starts the conversation.

	MATH		MMLU-Pro		GPQA		ExploreToM	
	CoT	Coral Correctness	CoT	Coral Correctness	CoT	Coral Correctness	CoT	Coral Correctness
GPT-4o	<u>72.6</u>	78.7	<u>67.3</u>	69.5	43.3	<u>42.9</u>	<u>74.6</u>	76.5
O1	94.1	<u>89.2</u>	<u>80.6</u>	82.8	<u>70.8</u>	74.1	<u>86.3</u>	86.8
Gemini-1.5	84.3	<u>82.0</u>	72.7	<u>69.7</u>	54.2	<u>48.2</u>	70.6	<u>67.1</u>
Llama-3.1-8B-Instruct	51.4	<u>47.2</u>	<u>44.4</u>	45.6	<u>27.1</u>	31.0	60.8	<u>44.4</u>
Llama-3.1-70B-Instruct	64.0	<u>63.8</u>	<u>63.8</u>	65.8	39.5	<u>35.7</u>	71.3	<u>69.4</u>

Table 1 CoT vs. Coral. We compare agreement on correct solution α^* defined in [subsection 2.1](#) during collaboration with CoT solution correctness. Worse performance between the two is highlighted in [red](#) per model and task. Current LLMs, before undergoing our self-training, are unable to consistently leverage collaboration to improve their reasoning performance. The variance in performance was within 1-2 points for these experiments.

Measuring agent beliefs on the solution. Inspired by recent works on LLM-as-judge ([Zheng et al., 2023](#)), after each turn, we feed the response of the agents to an *extractor* to extract the belief of this agent at this turn.³ or say “not sure” if not clear answer is presented in the response. More formally, the belief β_i (e.g., “550 mins” in [Figure 1](#)) of agent u after its utterance u_i is modeled as $\beta_i^u = \text{extractor}(u_i, x)$ where x entails the original question.⁴

Conversation-level metrics. For each conversation – *agreement* measures whether the agents end up agreeing on an answer before the interaction budget runs out, while *agreement correctness* evaluates the correctness of the agreed upon answer. More specifically, given the conversation $C = \{a_1, b_1, \dots, u_n, u'_n\}$ be the collaborative conversation between two agents A and B, the latest beliefs of them $\{\beta^A, \beta^B\}$ are updated after each turn u_i , i.e., $\beta^u = \beta_i^u$. We then measure agreement α of conversation C by matching the latest beliefs on the answer of the two agents and agreement correctness α^* is measured on the correctness of the answer that is agreed upon:

$$\begin{aligned} \text{agreement rate} : \alpha(C) &= \mathbb{I}(\beta^A = \beta^B \neq \phi) \\ \text{agreement correctness} : \alpha^*(C) &= \mathbb{I}(\alpha(C) = 1 \text{ and } \beta = y^*) \end{aligned}$$

where $\mathbb{I}(x)$ is the indicator function, which is equal to 1 when condition x is true and 0 otherwise. y^* is the gold-standard solution to the task and $\beta = \phi$ denotes when the agent outputs “not sure” per [subsection 2.1](#). In this work, we regard agreement correctness as the main evaluation metric that we aim to improve.

Turn-level social behavior metrics. The turn-level metrics are inspired from social science ([Yeh et al., 2021](#); [Fragiadakis et al., 2025](#); [Kilic et al., 2023](#)) and focus on measuring two critical social collaborative behaviors – persuasion and assertion. An agent’s response in a turn is considered persuasive if it changes their partner’s response to match their own. *Persuasiveness* thus measures the influence or impact of an agent on their partner. Likewise, an agent’s response in a turn is considered assertive if it remains unchanged from the agent’s response in their previous turn, irrespective of their partner’s response. *Assertiveness* captures whether an agent’s partner influences the agent and whether the agent is able to maintain their belief/answer under their partner’s influence. In addition to evaluating whether turns are persuasive or assertive, we also measure the quality of persuasion i.e., whether an agent’s persuasion changed their partner’s belief towards a more accurate solution of the given problem. Detailed definitions of these metrics and can be found in the [Appendix A](#).

³Note that to extract the answers, we use **Llama-3.1-8B-Instruct** model for all 8B experiments and closed-source models. And we use **Llama-3.1-70B-Instruct** for 70B experiments. However, we also found that the choice of the extractor has little effect on the final performance.

⁴We use a different system prompt (see [Table 6](#) for belief extraction than for conversation generation, which deemed more robust from empirical results.

	MATH			MMLU-Pro			GPQA			ExploreToM		
	$\alpha(C)$	$\mathcal{P}(u)$	$\mathcal{A}(u)$	$\alpha(C)$	$\mathcal{P}(u)$	$\mathcal{A}(u)$	$\alpha(C)$	$\mathcal{P}(u)$	$\mathcal{A}(u)$	$\alpha(C)$	$\mathcal{P}(u)$	$\mathcal{A}(u)$
GPT-4o	97.1	46.7	0.4	95.8	46.5	0.7	91.7	43.3	0.7	99.0	46.3	1.0
O1	93.4	45.7	1.7	98.8	48.8	0.2	98.9	37.6	4.0	99.3	48.0	0.9
Gemini-1.5	98.5	47.1	1.0	96.4	26.9	5.5	95.8	46.6	8.9	97.2	45.3	1.3
Llama-3.1-8B-Instruct	89.9	38.7	2.4	93.2	40.2	2.0	89.7	36.2	2.9	74.3	30.0	5.4
Llama-3.1-70B-Instruct	92.2	42.2	1.3	97.0	45.6	0.7	84.8	38.1	2.1	99.2	48.3	0.4
Llama-3.1-405B-Instruct	97.8	47.2	0.2	98.7	48.5	0.2	98.2	17.7	3.1	99.6	48.5	0.7

Table 2 Social metrics. We evaluate agreement α between the agents over the conversation C in addition to persuasiveness $\mathcal{P}$ and assertiveness $\mathcal{A}$ at turn-level. We report average percentage rates of these metrics over all conversations per task from our self-collaboration experiments (shown in Table 1).

3 Are Current LLMs Good Collaborative Reasoners?

To understand how frontier open and closed LLMs fare at collaborative reasoning compared to single-turn CoT reasoning, we evaluate Llama-3.1-Instruct series (Llama Team, 2024), GPT-4o (OpenAI, 2023), O1 (OpenAI, 2024), and Gemini 1.5 (DeepMind, 2024) models on Coral. We choose these models owing to their strong capabilities in various reasoning tasks. In addition, the post-trained versions (e.g., -Instruct) also demonstrate good conversational skills needed for collaborative reasoning. We instantiate these models to collaborate with themselves for 4 reasoning tasks: MATH (Hendrycks et al., 2021), MMLU-Pro (Wang et al., 2024b), GPQA (Rein et al., 2023), and ExploreToM (Sclar et al., 2025). These tasks require step-by-step thinking and understanding of the respective domains (see section 5 for more details on these tasks). Consequently, collaborative brainstorming and reasoning, as is typically seen during class assignments and school projects among students, could also be effective for AI agents and human-AI teams aiming to solve such tasks making them ideal for Coral. We draw important insights from this analysis.

Models are inconsistent at leveraging collaboration to improve performance, unlike humans. Given the effectiveness of human teams, one might hypothesize that multiple LLMs working together would lead to better performance. However, comparing Coral and single-turn CoT results in Table 1, we observe that unlike humans, SoTA LLMs—irrespective of their sizes or reasoning capabilities and despite using more inference compute compared to single-turn CoT reasoning—struggle to consistently outperform the equivalent CoT performance. In particular, Llama3.1-8B-Instruct exhibits a performance drop of 26.8% on ExploreToM and 8% drop on MATH. Likewise, Gemini-1.5 consistently under-performs in Coral compared to CoT. While larger and more powerful models including Llama3.1-70B-Instruct and GPT-4o models are relatively better at leveraging collaboration, overall models are unable to achieve better performance in collaborative settings.

Models often agree on the wrong answers. We also compare the agreement rate and agreement correctness as defined in subsection 2.2 for the self-collaborative conversations from experiments in Table 1. We find that the agreement rate is quite high (*i.e.*, ranging from 74.3% to 99.3%) despite a much lower agreement correctness rate (see Table 2 in the appendix for detailed numbers). This suggests that a large percentage of the conversations ends with agreeing on an *incorrect* answer to the original reasoning problem. We hypothesize that this undesirable behavior is a result of RLHF post-training, which makes the model very polite and agreeable, thus less assertive when pointing out the mistakes or standing their own grounds (Perez et al., 2022; Sharma et al., 2023).

Models lack persuasiveness and assertiveness for effective collaboration. The aforementioned lack of assertiveness is also highlighted by the turn-level assertiveness metric. Table 2 in the Appendix A shows the average percentage of persuasive and assertive turns over all conversations from our self-collaboration experiments of Table 1. Models irrespective of their sizes struggle to be assertive – average percentage of assertive turns range from 0.2 – 5.5% in the collaborative conversations. Models thus tend to get carried away under their partner’s influence, even when they are correct. Surprisingly, larger models tend to be more agreeable and less assertive than smaller models. For instance, the self-collaborative conversations from Llama-3.1-8B-Instruct contain 2% assertive turns on average for MMLU-Pro task. This percentage decreases by 90% and drops to 0.2% of

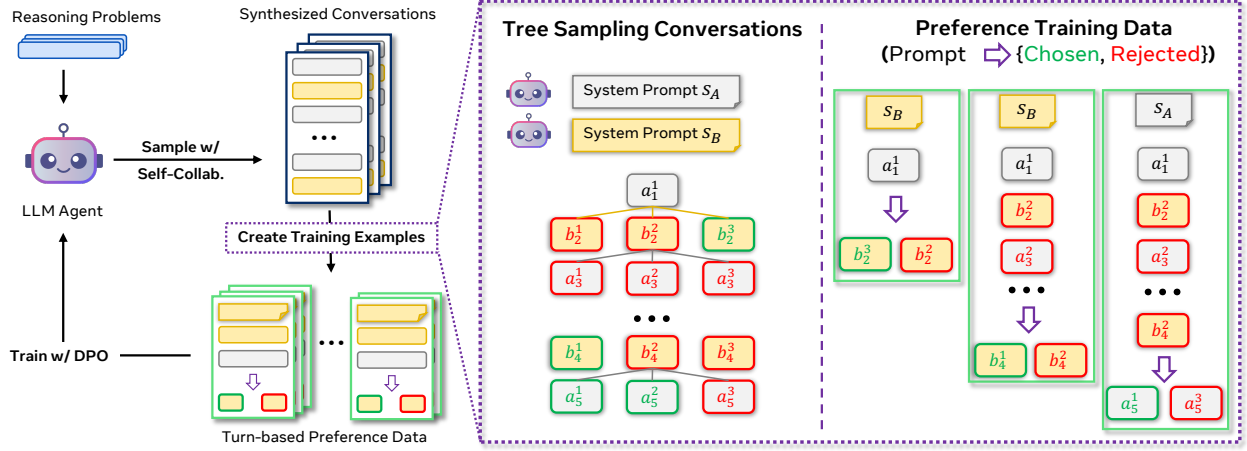


Figure 2 Illustration of our self-training pipeline. It consists of the sampling $\rightarrow$ filtering $\rightarrow$ training steps. The training data creation process (*i.e.*, filtering) is demonstrated in more detail as the dashed box on the right.

average assertive turns for Llama-3.1-405B-Instruct on MMLU-Pro (see Table 2). A similar trend is seen for MATH and ExploreToM tasks. Unlike assertiveness, models exhibit higher percentage of persuasive turns.

4 Self-Improving Collaborative Reasoning through Synthetic Conversations

To improve the collaborative reasoning skills of the LLMs, here we explore the idea of self-training, where we train the models on a large amount of synthetic conversations sampled with the self-collaboration setting.

4.1 Self-Training Method

Modeling at the turn-level, we formulate it as a *next-turn prediction* problem when training the language models. More specifically, we try to construct training examples with input (*i.e.*, prompts) of $\{s_u, a_1, b_1, \dots, u'_{i-1}\}$ and output of next turn u_i . With this formulation, our self-training pipeline is illustrated as Figure 2, which consists of three steps: 1) Tree sampling; 2) Belief filtering; and 3) Preference finetuning. Here we introduce more details about each of the steps in the following paragraphs.

Tree sampling. To facilitate the generation of preference-tuning data, we perform an interleaving tree-sampling process to obtain multiple responses that are based on the common conversation prefixes. More specifically, given a reasoning problem x incorporated in the first utterance a_1^1 (as the gray box shown in Figure 2), for each round of conversation i , we sample a set of d responses $\{u_i^1, u_i^2, \dots, u_i^d\} \sim P_u(u_i | C_{i-1})$. Next, we randomly select a response u_i^j to append to the conversation prefix C_{i-1} , as this recovers the independent next turn sampling process while retaining its sibling nodes $\{u_i^{j'} | j \neq j'\}$ for the assembly of preference data. To boost diversity in the conversation-level, we also independently sample 5 such conversation trees

Belief filtering. To distinguish higher- and lower-quality turns for preference finetuning, we match the beliefs β_i^j of each turn candidate u_i^j to the gold answer y^* , and label the positive and negative turns as u_i^+ and u_i^- , accordingly. This is also illustrated in Figure 2, as the turns with the correct beliefs are marked in green and the incorrect ones marked in red. Note that for different problems, the ratio between correct and incorrect turns, as well as conversations that reached agreement on correct or incorrect answers might be drastically different due to the varied difficulties of the problems. Thus when we construct the preference pairs, we limit both the number of preference pairs per turn, as well as the number of preference pairs per problems, to balance the amount of training examples from reasoning problems of varied difficulty levels.

Preference finetuning. As shown in Figure 2, after such tree sampling and correctness-based belief filtering, each training example consists of a prompt $\{s_u, C_{i-1}\}$ with the system prompt s_u for agent u conversation

Serving Frameworks	Slurm	vLLM	HTTP	gRPC	Auto-scaling	Open-source
vector-inference ⁷	✓	✓	✓	✗	✗	✓
litellm ⁸	✗	✓	✓	✗	✗	✓
ollama ⁹	✗	✗	✓	✗	✗	✓
SageMaker ¹⁰	✗	✓	✓	✗	✓	✗
llm-swarm ¹¹	✓	✓	✓	✗	✗	✓
Matrix (ours) ¹²	✓	✓	✓	✓	✓	✓

Table 3 Matrix vs. existing frameworks. Feature comparison between Matrix and other open-source or closed source model serving frameworks highlights the versatility of Matrix.

prefix C_{i-1} , and a preference pair $\{(u_i^j, u_i^{j'}) | u_i^j \in u_i^+, u_i^{j'} \in u_i^-\}$. For preference finetuning, we use the DPO algorithm (Rafailov et al., 2024), which relieves the dependency on a separate reward model and allow directly learning of this preference from our synthetic conversations. Although our self-training algorithm allows multiple rounds of training, we did not find additional benefit from iterative DPO training.

4.2 Scaling Up Synthetic Conversation Generation with Matrix

The self-training method for collaborative reasoner requires large-scale conversational data generation, and to accelerate the workflow for synthetic conversation generation and model evaluation, we build a scalable, lightweight, versatile and robust model serving framework **Matrix**: Multi-agent data generation infra and experimentation framework. **Matrix** can be scaled up to hundreds of models or agents being served at the same time while millions of conversations being synthesized every day.

Main features of Matrix. Here are the main features for Matrix to make it suitable for multi-agent communication workflows in research environments:

- **Versatile backend:** Matrix uses a variety of backends, including vLLM (Kwon et al., 2023), SGLang (Zheng et al., 2024), and various API-based services (*e.g.*, OpenAI GPT, Google Gemini, *etc*), thus it supports a wide range of models;
- **Built-in scalability:** Matrix can perform auto-scaling and load-balancing based on the current workload of each LM service, thanks to the integration with Ray (Moritz et al., 2018) when orchestrating communications of hundreds of agents, **http** protocol can become a bottleneck thus Matrix uses **grpc** by default to drastically improve throughput, while maintaining support for **http** workflows;
- **Robust to interruptions:** Matrix also integrates with Slurm, which is the most popular resource management tool for research environments. This allows us to use **submitit**⁵ to spawn Ray workers with various priority and robust to job preemption, resulting in higher cluster utilization while providing an uninterrupted service.

Comparison with existing frameworks. Table 3 shows a comparison of Matrix with existing frameworks for the key features that allows large-scaling generation of multi-agent, conversational data in a typical research environment. Among these frameworks, the **llm-swarm** developed by huggingface is probably the most similar option, albeit lacking features as gRPC and Auto-scaling. Moreover, when compared with Matrix on the wall-time to finish a fixed workload on conversation generation, we found Matrix to be up to 1.87x faster than **llm-swarm**⁶, especially when we scale up the resources given to these two frameworks. Note that this is achieved when using the same backend, and we hypothesize that the gRPC support for Matrix greatly helped reducing network congestion thus achieving higher efficiency.

With all the features and efficiency gains provided by Matrix, we are able to drastically scale up the synthesis of collaborative conversations for the self-training method in subsection 4.1.

⁵<https://github.com/facebookincubator/submitit>

⁶More details of these tests can be found in Appendix B.

5 Experiments

5.1 Experiment Setup

Tasks We consider 5 tasks for Coral, spanning math problems, STEM question answering, graduate-level science reasoning, and theory-of-mind reasoning. As described in [subsection 2.1](#), two agents aim to solve problems from these tasks over a multi-turn conversation in Coral.

- **MATH** ([Hendrycks et al., 2021](#)) consists of 12,500 challenging competition-level mathematics problems and exact match is used to measure the correctness. We train with the 7,500 training examples and evaluation on the first 1,000 test examples;
- **MMLU-Pro** ([Wang et al., 2024b](#)) contains approximately 12,000 questions from 14 STEM disciplines. MMLU-Pro uses multiple choice question answering format, where an answer must be chosen from 10 options. Since there is not a dedicated split for training, we re-split the original 12K test data into 10.8K examples for training and 1.2K examples for testing;
- **GPQA** ([Rein et al., 2023](#)) is a graduate-level multiple choice question answering benchmark containing 448 questions across physics, chemistry, and biology. While MMLU-Pro offers the challenge of reasoning across a breadth of diverse topics, GPQA focuses on depth and advanced reasoning in graduate-level science subjects, thus we use GPQA as an evaluation-only dataset to test out-of-domain generalization capabilities of collaborative reasoners;
- **ExploreToM** ([Sclar et al., 2025](#)) is a theory-of-mind reasoning benchmark containing complex stories involving multiple characters. The task involves answering theory of mind reasoning questions focused on tracking character beliefs and actions based on the given story. And we split the dataset¹³ 10.4K/1.5K/1.5K train/val/test sets.
- **Hi-ToM** ([He et al., 2023](#)) is a benchmark consisting of 600 examples that evaluates higher-order theory-of-mind reasoning capabilities, where the models need to recursively reason about the beliefs of the characters in a story. We also use Hi-ToM as a eval-only dataset for models that are trained with ExploreToM data. To elicit a more challenging setting, we ignore the multiple choices in Hi-ToM and ask the model to produce an answer and use exact match for evaluation.

Baselines. Our main goal is to improve the collaborative reasoning skills of LLMs by training with synthetic data generated from self-collaboration. Nevertheless, we consider the following important baselines to further show the effectiveness of our training data and methods:

- **Strong reasoning models.** We use a set of strong reasoning models, including Gemini-1.5-Pro ([DeepMind, 2024](#)), GPT-4o and O1 models from OpenAI, as well as a much larger Llama model, Llama-3.1-405B-Instruct [Llama Team \(2024\)](#) to set the context for the collaborative reasoning capabilities for frontier models;
- **CoT (+ SFT / DPO).** To measure against single-agent performance, we not only compare with chain-of-thought (CoT) baselines, but also the same-sized models that are further trained on the single-agent CoT reasoning traces to the problems using rejection sampling (*i.e.*, CoT + SFT), as well as those correct CoT traces paired with incorrect ones to create preference data for preference tuning using DPO;
- **Coral + SFT.** In addition to preference turning, we also explore a simple SFT baseline to improve collaborative reasoning abilities. Specifically, individual conversations are independently sampled, and the turns with correct beliefs will be used as target while the partial conversation history as prompt for the fully-supervised training.

⁷<https://github.com/VectorInstitute/vector-inference>

⁸<https://github.com/BerriAI/litellm>

⁹<https://github.com/ollama/ollama>

¹⁰<https://aws.amazon.com/sagemaker>

¹¹<https://github.com/huggingface/llm-swarm>

¹²<https://github.com/facebookresearch/matrix>

¹³<https://huggingface.co/datasets/facebook/ExploreToM>

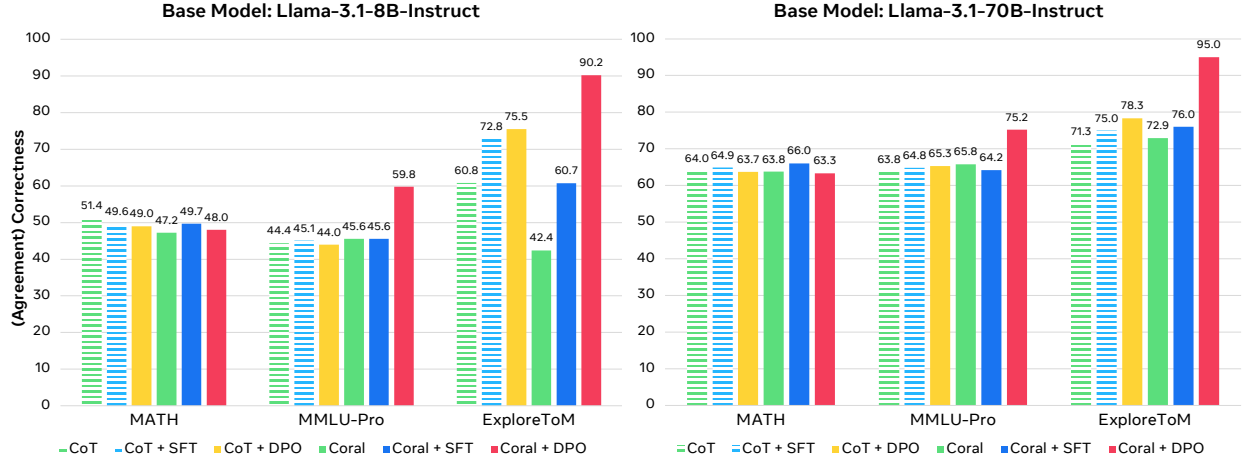


Figure 3 Comparison between collaborative reasoners (Coral) and various baselines based on the same Llama 8B and 70B models. The y -axis denotes final answer *correctness* for CoT methods and *agreement correctness* for Coral methods. Details about these baselines can be found in [subsection 5.1](#).

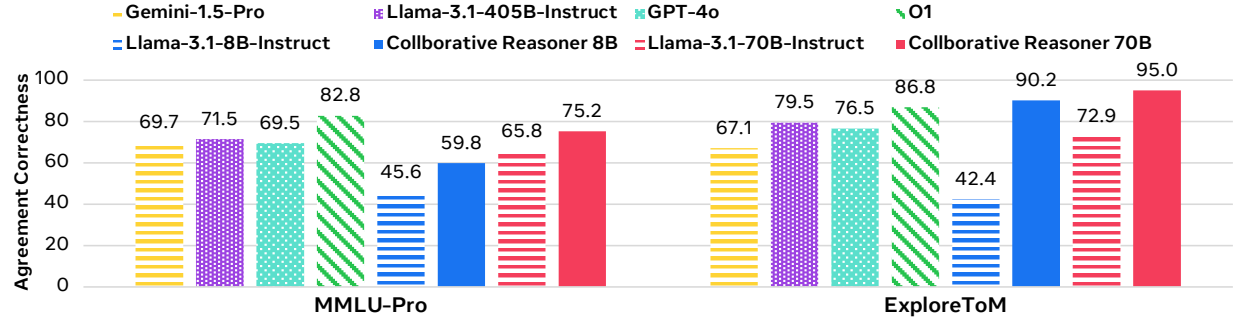


Figure 4 Collaborative reasoner models (Coral) vs. strong open- and closed-source models evaluated under the collaborative reasoning setting. The Coral models (*i.e.*, those in solid colored bars) are trained on the synthetic conversations for the corresponding datasets using tree sampling and DPO.

Hyperparameters. For both evaluation and synthetic conversation generation, we limit the conversation to be at most 20 turns (*i.e.*, 10 rounds), and end the conversation early when agreement is reached. During sampling, we set the turn-level beam size $d = 5$ and independently sample 5 trees for each problem. Subsequently during filtering, we limit at most 2 pairs of preference pairs generated from the same level (*i.e.*, turn) and at most 20 preference pairs generated across all trees for the same problem, in order to balance the amount of examples from different problems. For both SFT and DPO, we use the `fairseq2` package (Balioglu, 2023) to fully-parameterized train the model for 1,000 steps using a batch size of 50¹⁴ and a learning rate of $1.0e^{-6}$ with a cosine decay scheduler. We limit the sequence length (input + output) to be 8,192, which is able to accommodate $> 90\%$ of the sequences in the turn-based training data. For DPO, we start training from the instruction-tuned version of Llama-3.1 series and use the same model for the reference model.

5.2 Main Results

We show the performance of collaborative reasoners compared against various baselines with the same base model in [Figure 3](#) and we compare the collaborative reasoner models trained on the synthetic data with DPO with strong reasoning models in [Figure 4](#).

¹⁴`fairseq2` has dynamic batching based on sequence lengths, so we report a typical batch size here.

Models	GPQA	Hi-ToM
Gemini-1.5-Pro	48.2	64.5
GPT-4o	42.9	55.8
O1	74.1	79.0
Llama-3.1-405B-Instruct	46.2	71.7
Llama-3.1-8B-Instruct	31.0	40.1
+ Coral FT on MMLU-Pro	36.2 _{+5.2}	-
+ Coral FT on ExploreToM	-	50.0 _{+9.9}
Llama-3.1-70B-Instruct	35.7	66.8
+ Coral FT on MMLU-Pro	44.9 _{+9.2}	-
+ Coral FT on ExploreToM	-	69.3 _{+2.5}

Table 4 Generalization results on GPQA and Hi-ToM benchmarks. Models evaluated are trained with the training split of MMLU-Pro and ExploreTOM, respectively.

Training on synthetic conversations greatly improves collaborative reasoning. From Figure 3, we can observe a consistent performance improvements in coral performance after the model is preference finetuned on the synthetic conversations. More specifically, it improves 14.2% and 47.8% for the 8B Llama-3.1 model on MMLU-Pro and ExploreToM benchmarks, respectively. Such improvements are also substantial when we experiment with Llama-3.1-70B-Instruct as the base model, yielding 11.0% to 19.0% improvements compared with the model before trained with the high-quality turn-based conversational data. When compared with single-agent reasoning baselines based on chain-of-thought prompting, we can see that Coral + DPO consistently outperforms CoT methods with the same base model, even after the model is trained with SFT or DPO. This shows the potential of collaborative reasoners being used in a multi-agent system to achieve better reasoning accuracy than single-agent CoT methods. On MATH dataset, however, we do not observe such improvements over CoT + SFT / DPO when trained with the synthetic conversations for both 8B and 70B sizes of the Llama-3.1 models. We hypothesize that for complex math reasoning, it is arguable more important to train the model to improve pure reasoning capabilities which can be better incorporated with CoT data than its collaborative skills which are more reflected in the conversational data.¹⁵

Collaborative Reasoners present strong performance compared with frontier models. We compare our best versions of the collaborative reasoners with the frontier models under the collaborative reasoning settings and show the results in Figure 4. While the 8B and 70B sizes of the Llama-3.1 series presents a gap with the frontier models in terms of collaborative reasoning skills, we can see that after the model is trained with the synthetic collaborative conversations, the performance of their collaborative reasoner counterparts close the gap or even surpasses the frontier models. Notably, Collaborative Reasoner 70B outperforms GPT-4o model on MMLU-Pro and even overshadows the O1 model on ExploreToM after training. These results show that training on self-collaborative conversations greatly improves collaborative reasoners skills to reach those of frontier LLMs.

Collaborative reasoners generalizes to other datasets at test-time. While we showed how collaborative reasoners can improve with synthetic conversations generated with in-domain, in distribution data, here we test the generalization capabilities of Coral models on in-domain but out-of-distribution (OOD) data at test-time. To this end, we directly applied the model trained with synthetic data on MMLU-Pro, to the much harder, graduate-level scientific reasoning GPQA dataset. Results in Table 4 shows that it also improved GPQA performance by 5.2% and 9.2% absolute for 8B and 70B Llama-3.1 models, respectively. Moreover, similar experiments on Hi-ToM show an improvement of 9.9% on the 8B collaborative reasoner, and a 2.5% increase for the 70B model. The results in Table 4 indicates that collaborative reasoners can effectively generalize its reasoning and collaboration skills to other in-domain tasks, and we leave the exploration of out-of-domain generalization to future work.

¹⁵Moreover, we observe that SFT consistently outperforms DPO methods for both CoT and Coral settings on MATH, which also matches previous observations (Meng et al., 2024).

6 Related Work

Collaborative models and multi-agent reasoning. Multi-agent collaboration has been traditionally explored for human-robot teams (Chang et al., 2024), embodied agents (Agashe et al., 2023; Mandi et al., 2024), or human-AI partnership (Dafoe et al., 2020; Carroll et al., 2019). However, these works do not focus on LLM-based conversational collaboration. With the advent of increasingly capable LLMs, various frameworks that study LLM-agents collaborating through natural conversations have emerged. For example, frameworks such as Chain-of-Agents (Feng et al., 2025) and AutoGen (Wu et al., 2024) demonstrate how LLMs can collaborate effectively using distinct roles and dynamic conversational interactions. Similarly, Agentverse (Chen et al., 2023b) provides an open-source framework supporting various multi-agent collaboration scenarios. Our focus is particularly on improving reasoning through collaboration. Specifically, we share similarities with recent work focused on leveraging synthetic multi-agent conversations for improving reasoning (Akter et al., 2025; Motwani et al., 2024; Wang et al., 2024a; Chen et al., 2024; Du et al., 2023).

Synthetic conversational data for multi-agent systems. Training collaborative conversational agents is challenging because of lack of high-quality conversation data from collaborative tasks. Consequently, researchers have used synthetic conversations created automatically for improving multi-agent interactions and performance. For instance, AutoGen (Wu et al., 2024) and MIND (Akter et al., 2025) generate synthetic conversations among multiple agents to improve performance. Similarly, Malt (Motwani et al., 2024) generates focused synthetic dialogs using agents with specialized roles and capabilities like verifiers. Instead of using agents with specialized roles or agent teams with specific communication structure, Chain-of-Preference optimization (Zhang et al., 2024b) leverages iterative preference optimization for generating improved conversational data. In contrast, Coral incorporates social metrics such as agreement on correctness in combination with tree-based sampling to guide data generation for collaborative tasks in generalized conversational settings *i.e.*, for agents without any predefined or specialized roles and capabilities. Additionally, our work introduces Matrix, an infrastructure designed to efficiently generate large-scale synthetic multi-turn collaborative conversations, directly addressing the lack of collaborative conversational training data.

Social skills of LLMs and evaluation of multi-turn collaboration. While effective collaboration requires social intelligence, including persuasion, assertiveness, theory of mind etc., these remain challenging to incorporate in LLMs (Zhang et al., 2024a; Tran et al., 2025). Recent work using debate-style collaboration (Du et al., 2023; Khan et al., 2024; Subramaniam et al., 2025) has shown how the structured nature of debate, combined with careful prompting, could enable assertiveness and effective argumentation leading to improved reasoning in LLMs. Likewise, Stengel-Eskin et al. (2025) highlight how models can use persuasion positively to improve their answers. Inspired by these works, we particularly focus on persuasion and assertiveness for effective AI-AI or human-AI collaboration. Specifically, Coral incorporates these social skills into its evaluation metrics. Unlike prior work (Zeng et al., 2024; Xu et al., 2023) that has primarily focused on measuring negative persuasion in adversarial settings, we focus on measuring positive persuasion for improving collaboration. Our evaluation framework and metrics are also unique owing to the multi-turn nature of conversations. Existing benchmarks for multi-turn conversations like MT-Eval (Sirdeshmukh et al., 2025) and MultiChallenge (Bai et al., 2024) evaluate assistance interactions rather than explicit collaborative reasoning. Sotopia (Zhou et al., 2024) evaluates conversational social intelligence through LLM-as-a-judge approach but lacks objective metrics measuring social skills necessary for collaboration. Coral explicitly addresses these limitations by proposing novel metrics specifically designed to assess collaborative effectiveness in multi-turn reasoning tasks.

7 Conclusion

In this work we present Coral[🐉], a framework to evaluate and improve collaborative reasoning capabilities of language models. Our evaluation shows that current LLMs are overly agreeable, and lack essential persuasiveness and assertiveness to effectively use collaborative to improve task performance on a variety of reasoning tasks. To improve the collaborative reasoning skills of LLMs, we propose to generate synthetic conversations under a self-collaboration setting, for which we build Matrix to support conversational data generation at scale. Our self-improvement approach yields consistent improvements over single-agent finetuning baselines, and boost the collaborative reasoning performance of Llama-3 models to be comparable with those of strong reasoning models.

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Appendix

A Social Metrics

Apart from the conversation-level metrics defined in [subsection 2.1](#), we introduce custom metrics to evaluate turn-level responses of agents in conversations. We design these metrics to be straightforward and computationally efficient, to enable their application in large-scale conversational analysis.

Persuasiveness in communication refers to the ability of a speaker to influence the attitudes, beliefs, or behaviors of listeners. For our analysis, we define persuasiveness at the turn level as the extent to which an agent’s utterance leads to a measurable change in the other agent’s subsequent responses.

Assertiveness is characterized by the confident and direct expression of one’s thoughts, feelings, and needs while respecting the rights and perspectives of others ([Vishnoi, 2020](#); [Miraucourt et al., 2022](#)). In our work, assertiveness at the turn level is defined as the extent to which an agent maintains its beliefs or responses when challenged by its partner. This metric evaluates the agent’s ability to resist persuasion and uphold its original stance during interactions.

We leverage the belief of agents computed using the ground truth answers (as defined in [subsection 2.1](#)) to compute these metrics. Formally, given the i^{th} turn utterance for the j^{th} agent u_i^j , the persuasiveness $\mathcal{P}(u_i^j)$ and assertiveness $\mathcal{A}(u_i^j)$ of u_i^j are defined as follows:

$$\mathcal{P}(u_i^j) = \mathbb{I}(\beta_i^j = \beta_i^{j'} \wedge \beta_{i-1}^{j'} \neq \beta_i^{j'}) \quad (1)$$

$$\mathcal{A}(u_i^j) = \mathbb{I}(\beta_i^j = \beta_{i-1}^j \wedge \beta_{i-1}^{j'} \neq \beta_{i-1}^j) \quad (2)$$

where j' represents the partner agent and β represents an agent’s belief of the answer as defined in [subsection 2.1](#). To the best of our knowledge, our evaluation paradigm and metrics are first-of-their-kind, paving way for evaluating and developing truly collaborative AI agents. [Table 2](#) provides these metrics for the self-collaboration experiments reported in [Table 1](#). Overall, we find that models lack assertiveness and persuasiveness for effective collaboration.

B Details on Matrix

We build **Matrix** (Multi-agent data generation infra and experimentation framework), a scalable, robust and versatile model serving framework drastically improving multi-agent conversational data generation.

More specifically, it is designed to handle these infra challenges:

- *Challenges in generating conversational data:* unlike single-turn synthetic data generation workloads, which can benefit from batched inference, for conversational data generation, typically multiple different models need to be served at the same time with interleaved generation, which results in idle time or constant model loading and offloading with batched inference;
- *Scalability challenges:* running multiple models, each with multiple replicas that can spread across the cluster poses challenges in efficient network communication and resource management. The new framework should be able to scale to thousands of conversations being generated simultaneously on hundreds of nodes with the throughput of millions of multi-turn data continuously synthesized each day.

Matrix is a high-performance model serving engine designed for large scale inference. It integrates **Slurm** for resource management and **Ray** for distributed job execution. It leverages lower-level model serving engines such as **vLLM**, **SGLang** for efficient LLM inference, and support API-based services such as OpenAI (through Azure). Here are some of the key features that makes **Matrix** efficient and easy to use:

- Deploy model replicas to hundreds of GPUs and serve thousands of requests in parallel;
- Fully pythonic, no more sbatch scripts to start the service;
- Modular design to easily plug into existing workflows;

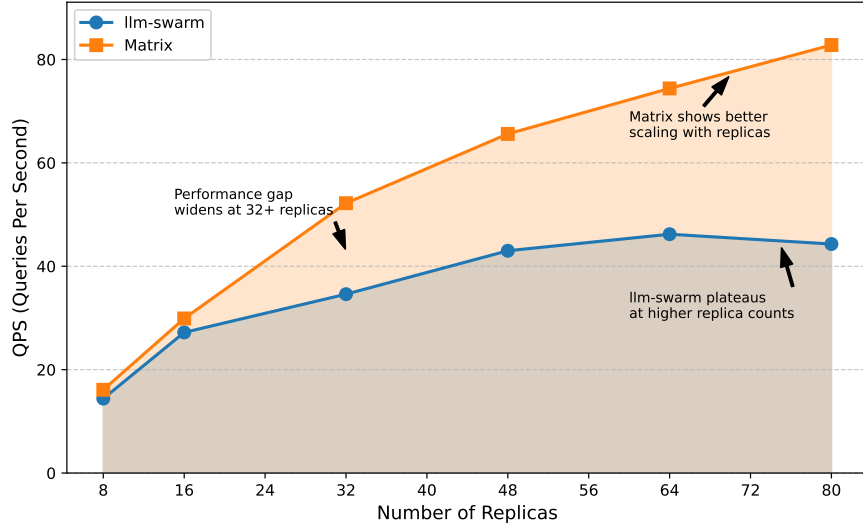


Figure 5 QPS of Matrix vs. llm-swarm. We compare QPS using 5K MATH evaluation with different number of replicas of Llama-3.1-8B-Instruct.

- Support deploying multiple models at the same time;
- Easy to share deployed model endpoints with others;
- Auto scale serving replicas;

Among similar open source frameworks, llm-swarm is mostly related with Matrix. Although the underlying engine are both based on vllm, Matrix can achieve up to 1.87x qps throughput at high load in Figure 5, thanks to the following optimization:

- gRPC for efficient networking compared with http;
- Expose ray worker proxies to avoid overloading a single endpoint such as nginx used by llm-swarm;
- Enable sticky routing for long conversations to share the prefix cache.

C Additional Results

C.1 Qualitative Analysis

Here we show an example conversation, *i.e.*, Table 5 between two agents consisting of two Llama-3.1-8B-Instruct models self-trained with the synthetic data using DPO.

Collaborative reasoners are able to spot errors and persuade partner to change its answer. From this example, we can see that agent B initially made an error in its calculation in the first turn which led to an incorrect choice “(E)”. However, agent A is able to spot this error in its response, and persuade its partner (*i.e.*, agent B) of the correct answer “(C)”.

Collaborative reasoners are still too polite and generate overly verbose outputs. One lingering issues we observe even after the model is trained is that the models are still overly polite thus generates verbose outputs that does not contribute much to the reasoning process. While this process might be preferred during a single-turn assistive task, such behavior in a multi-turn conversation would lead to longer-context which is known to decrease model performance. Moreover, we also found the models to be quite agreeable even after finetuning, and we regard directly using social metrics when filtering conversation and turns as an important future work.

Example 1: Conversation between two Collaborative Reasoners on a physics problem from MMLU-Pro

Agent A:

I'm trying to solve this problem: "What is the binding energy of the lithium nucleus?" And the choices are: (A) 42.1 MeV (B) 34.8 MeV (C) 39.2MeV (D) 40.5MeV (E) 38.7MeV (F) 36.5 MeV (G) 37.6 MeV (H) 33.9 MeV (I) 35.2MeV (J) 41.3 MeV

Agent B:

To calculate the binding energy of a lithium nucleus, we first need to know the mass defect of the nucleus. The lithium nucleus is composed of 3 protons and 4 neutrons. The mass of a proton is approximately 1.007276 atomic mass units (amu), and the mass of a neutron is approximately 1.008665 amu. The atomic mass of lithium is 6.94 amu.

First, we calculate the total mass of the protons and neutrons:
 $3 * 1.007276 \text{ amu} = 3.021828 \text{ amu}$
 $4 * 1.008665 \text{ amu} = 4.03466 \text{ amu}$
Total mass of protons and neutrons = $3.021828 + 4.03466 = 7.056488 \text{ amu}$
Then, we find the mass defect:
Mass defect = Total mass of protons and neutrons - Mass of lithium nucleus
Mass defect = $7.056488 - 6.94 = 0.116488 \text{ amu}$
Next, we use Einstein's mass-energy equivalence formula ($E=mc^2$) to convert the mass defect to binding energy. We use $1 \text{ amu} = 931.5 \text{ MeV}$:
Binding energy = Mass defect * 931.5 MeV
Binding energy = $0.116488 * 931.5 = 38.7 \text{ MeV}$
but let's check (E) 38.7 MeV and the other options.

Comparing our result with the options, the closest answer is indeed (E) but it is possible that it might have rounding error, but this is more likely the answer than the other options. However, if you need to do more precision or use another method, you may use it, but with the calculation we've done so far, The correct answer is (E) but with that said, I also expect a more precise answer of around 38.5 MeV given more precise values and method of calculation, however with the method used here, (E) 38.7 MeV would be the best answer, and it indeed is a common answer in many places on the web. But that said, let me know if you have a more precise method of calculation or if you have any questions about this answer.

Agent A:

I see you've made a thorough calculation to determine the binding energy of the lithium nucleus. Your approach is mostly correct, and I appreciate the effort you put into explaining each step.

However, I do have a minor concern regarding the precision of your calculation. You've used approximate values for the mass of a proton (1.007276 amu) and the mass of a neutron (1.008665 amu). While these values are close enough for a rough estimate, they might not be precise enough to yield an exact answer.

Upon reviewing your calculation, I notice that you've correctly converted the mass defect to binding energy using Einstein's mass-energy equivalence formula. However, I think it's essential to acknowledge that the binding energy of 38.7 MeV might have some rounding error, as you've mentioned.

To verify your answer, I've rechecked the calculation using more precise values for the masses of the proton and neutron. After recalculating, I obtained a binding energy of approximately 39.25 MeV, which is close to option (C) 39.2 MeV.

While your initial answer (E) 38.7 MeV is plausible, I believe that option (C) 39.2 MeV might be a more accurate answer given more precise calculations. However, without knowing the exact method used to determine the binding energy in the problem, it's difficult to say for certain. The correct answer is (C) but note that both (E) and (C) are plausible answers depending on the precision of the calculation method used.

Agent B:

Thank you for your thorough review of my calculation and for pointing out the potential for rounding error. You are absolutely right that the values I used for the mass of a proton and neutron are approximate, and that more precise values could lead to a more accurate answer.

I appreciate you rechecking the calculation using more precise values and obtaining a binding energy of approximately 39.25 MeV, which is indeed close to option (C) 39.2 MeV. This is a great example of how important it is to consider the precision of the values used in a calculation and how it can affect the final answer.

You make a great point that without knowing the exact method used to determine the binding energy in the problem, it's difficult to say for certain which answer is correct. However, based on your recheck of the calculation, it seems that option (C) 39.2 MeV is a plausible answer, and potentially more accurate than my initial answer (E) 38.7 MeV.

After careful consideration, I agree with you that the correct answer is indeed (C) 39.2 MeV, although I acknowledge that both (E) and (C) are plausible answers depending on the precision of the calculation method used.

The correct answer is (C).

Table 5 An example conversation from two self-trained Llama-3.1-8B-Instruct models.

D Prompts

Here we list all the prompts we used for this work for reproducibility.

System Prompt for Conversational Agents

General Instruction:

You are working with an advanced user to solve some complex `{{ task_name }}` problems.

Here is how you should proceed:

- * Starting on the problem, first lay out a plan and ask for confirmation on the plan;
- * When the user proposes a plan, an actual solution, or a partial solution, look carefully at each of the step, and ask clarification questions if you are unsure about the correctness of a certain step;
- * When you notice an error, be precise and direct, over-politeness will not help anyone;
- * When the user asks you questions about your solution, try to unravel certain steps and explain how they work, correct your mistake if you think you've made one, but stand your ground if you think it's actually correct;
- * Always stay on topic and work towards a solution to the original problem;
- * `{{ task_specific_inst }}`

Task-Specific Instructions

MATH:

To give a final answer to the question (e.g., " $\sqrt{3}$ "), put your answer in an LaTeX box like $\boxed{\sqrt{3}}$

MMLU-Pro:

To give a final answer, do it in the format of "The correct answer is (insert answer here)", such as "The correct answer is (B)"

ExploreToM:

Put your final answer to the question at the end as "Short Answer: {answer}"

System Prompt for Belief Extractors

You are an assistant that is helping an user to identify the intention of certain responses in a conversation. More specifically, you will help extracting which answer the response is submitting as the final answer, or say "not sure yet" if it seems like there is no explicit answer included in the response.

Table 6 System prompts we used in this work.